Nudge a local-AI assistance dog for deaf, blind and deafblind people

Agentverse 2026 · Robotics track · Unitree Go2 in MuJoCo · local Gemma 4 at its core · 100 % local AI

Safety always on.
Local Gemma for the unknown.

2 million+
people in the UK living with sight loss

18 million
adults deaf, with hearing loss or tinnitus: 1 in 3 (RNID)

450,000+
deafblind people in the UK, rising to 610,000+ by 2035 (Sense)

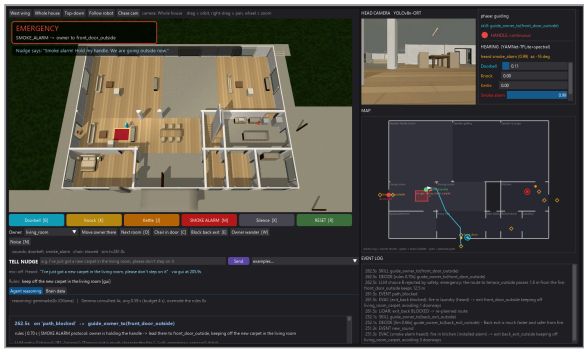
16–18 months
average wait for a guide dog; some wait 2–4 years

The problem

Deafblindness is the hardest case: over 450,000 people in the UK, rising to over 610,000 by 2035, and Sense found that 1 in 5 deafblind people received no social care at all. An assistance dog changes their life, but there are far too few: Guide Dogs UK has only around 3,500–4,800 working partnerships, people wait 16–18 months on average and some 2–4 years, and one guide dog is reported to cost £55,000 to over £100,000 from birth to retirement. A hearing dog costs about £40,000 over its life, takes about 1.5 years to train and has a waiting list of around two years. **A deaf person who can't hear the smoke alarm can't wait two years.**

Our user and what Nudge does

Nudge is a Unitree Go2 quadruped in a furnished house. It listens all the time with stereo ears and sorts household sounds by urgency: **emergency** (smoke alarm), **needs action** (kettle, oven timer), **someone wants you** (doorbell, knock). It finds the owner with its camera (and their voice when they speak), nudges them and alerts them the way their profile needs: haptic guide handle (deafblind), light bar and screen (deaf), speech only when close (blind). Then it guides them to the source, or out of the house in a fire.



Smoke alarm: the emergency pre-empts everything; LiDAR finds the back exit blocked, Nudge re-plans, and safety rejects a Gemma exit choice that passes near the fire. Perception (head camera, hearing) is shown next to the actions.

The main idea

Safety is always on; everything else is learned from the owner. The safety features (smoke-alarm pre-emption, the LLM watchdog, the speed cap near the owner, calling for help) are plain code that can't be switched off, overridden by the AI or downgraded by anything the owner says. On top of that, Nudge **remembers what the owner tells it and keeps doing it in the future**: rules ("don't step on the new carpet") and tasks ("every night at 10, take me to bed"), turned into structured rules and saved task programs by local Gemma, then enforced and repeated by plain code. **Gemma is the brain in between:** it understands the owner and decides what to do; plain code makes sure it is always safe.

System architecture

Two processes over ZeroMQ, so physics never waits for AI. **Simulator:** 500 Hz physics, head RGB-D camera (10 Hz), 36-ray 2D LiDAR, stereo 16 kHz ears with wall and distance effects. **Brain** in two layers: a **reflex layer** in plain code (smoke-alarm pre-emption, obstacle stops, speed cap near the owner, stale-command watchdog; milliseconds) and a **thinking layer** (Gemma, below), woken only on events, which picks one option from a menu of legal actions built from live perception, so it can't produce an action that isn't on the menu.

Gemma 4 at the core

Local **Gemma 4 E2B** (Ollama, on the laptop GPU) is the model we built Nudge around. At every event it **decides**: which room to search first, re-nudge or lead the owner out, which exit in a fire, whether to call 999, and it **explains** each decision in plain words shown on the screen. It turns the owner's **sentences into rules and task programs**, writes the short message Nudge says to the owner, **reasons about sounds it has no rule for** (glass breaking, a dog) and (built, off in the take) **looks at the owner through the camera** to spot someone lying on the floor that YOLO misses. Answers are JSON-schema constrained to the legal options. Measured on the recording laptop: **0.19 s** median per decision, **93 %** right on real logged decisions, 100 % inside the 4 s watchdog, so no fine-tuning was needed; every run is logged and an eval gate (90 %) re-checks Gemma after any prompt change.

Where each model runs (one laptop with an RTX 5060; no cloud calls in the loop)

Model	Job	Runs on	Latency
Gemma 4 E2B (Ollama)	**all decisions + explanations; owner's words → rules / tasks; camera check**	laptop GPU	0.19 s median (0.26 s p95) per decision; ~0.6 s per sentence
YAMNet + spectral rules	household sounds	CPU	alert 0.5 s after a sound starts (median)
YOLOv8n (ONNX)	owner + objects	CPU	real time at the camera's 10 Hz
Whisper base.en	owner's voice	CPU	under a second per sentence
Piper TTS	Nudge speaks	CPU	local, offline

Adapting when the scene changes

- The owner moves mid-run; a chair blocks a doorway, or the back exit in a fire, and the robot re-plans. It notices when it's stuck, backs off, remembers the obstacle and re-routes.
- A doorbell from the TV is recognised as the TV (0 of 243 real door sounds mistaken for the TV); a kettle during a doorbell mission is deferred, then handled; it never re-scans a room its camera already cleared.
- **Built and measured, off in the recorded take:** it maps a house by itself (99.8 % of the floor, walls 97 % precise, exits found within 0.01–0.21 m); it estimates its own position from LiDAR (~8 cm error vs 9.4 m drift from odometry alone; passes the demo story with it on); a second home (an apartment) runs with the same code.

Measured

- **Sounds:** 96 % of demo sounds detected in 1,000 random trials over 5 noise profiles; the smoke alarm was never missed; 0 % false alerts from background noise.
- **Owner detection:** seen in 97–98 % of head-camera views. **Gemma:** right on 93 % of real logged decisions, 100 % answered inside the 4 s watchdog.
- **Adaptability benchmark:** 9 of 12 randomised, never-seen situations passed (75 %). **Teaching by sentence:** 19/19 test sentences turned into the right rule or task, or refused.
- **Scenario tests (ground-truth judge):** default suite 14/14 PASS; robot-map mode 22/24 + 9/9 soak; apartment doorbell, smoke, kettle PASS.

Critical failure mode and how it's handled

The critical failure is **a missed or late smoke alarm, or not getting the owner out**. The alarm is checked first, in plain code, and never gated by other rules: it pre-empts any task in **0.6 s** with a continuous haptic alert. The exit choice is fire-aware (walk length vs distance from the fire, never through the fire room, exits the LiDAR sees blocked are dropped), and a Gemma exit choice that passes closer to the fire than the planner's is rejected. Nudge asks "Shall I call 999?" while still leading the owner out and calls on "yes" or no answer; if the owner isn't found within 40 s it calls anyway.

Safety

The LLM only chooses from legal options; a 4 s watchdog hands any slow or offline answer to deterministic rules; speed is capped at 0.4 m/s within 1 m of the owner; nothing the owner says can downgrade a smoke alarm; if the owner doesn't respond to repeated nudges, Nudge alerts an emergency contact and stays with them.

Limits (what it cannot do yet)

- **Sound:** the demo uses synthesized clips; real recordings are weaker (smoke 96 %, doorbell 64 %, worst under loud noise).
- **Knowing the house:** room names, doors and alarm locations come from a hand-written floor plan (the robot's own map gives walls and exits, not names); it only recognises furniture a standard detector knows.
- **Body:** walking is kinematic and the handle, lights and screen are simulated; self-localisation needs a known start.
- **Owner unreachable:** if the only way to the owner is blocked, it re-routes but doesn't yet escalate.
- **Next steps to a real Go2:** localisation on real sensors, real-room hearing, building the handle hardware.

Sources: RNID (prevalence of deafness and hearing loss); Guide Dogs UK / Royal Canin (sight loss, partnerships); Sense (deafblindness statistics, 2023; evidence to Parliament); Hearing Dogs for Deaf People; BBC, Guardian, ITV and Kent Online reporting on guide-dog waiting times and costs.

Team Nudge: Eden, Finn, Ricktho, Sean, Teo · Scene: nerv-world house (MIT), Go2 from MuJoCo Menagerie (BSD-3) · Numbers measured in simulation on 2026-11-08.